

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Analyzed and mitigated patient safety hazards within Medtronic software by collaborating with cross-functional teams, focusing on risk reduction for cardiac medical products.
- Developed and maintained risk management files in compliance with **ISO 13485** and **ISO 14971**, ensuring alignment with design inputs and outputs for released software.
- Executed reliability and safety engineering studies, including **design of experiments (DOE)**, to evaluate design concepts and assess system performance for medical devices.
- Supported the implementation of reliability and safety engineering programs to meet company, customer, and **global regulatory standards**.
- Provided data-driven risk–benefit insights and proposed innovative risk mitigation strategies to stakeholders, prioritizing patient safety in software design.
- Recommended and implemented design improvements and test methods using **statistical process control** techniques to enhance product reliability and safety.

SKILLS

Safety & Reliability – Hazard Analysis, Risk Management, ISO 13485, ISO 14971, IEC 62304, PHA, HAL, FTA, DFMEA, Design for Reliability (DfR), Design for Six Sigma (DFSS), Robust Design

Software Development – Java, Spring Boot, Python, C++, JavaScript, C#, .NET, HTML, Mobile Applications, REST APIs

Data Analysis & Tools – Power BI, Tableau, R/RStudio, pandas, NumPy, scikit-learn, SQL, DAX, Power Query

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git, CI/CD

Methodologies – Agile, Scrum, Design of Experiments (DOE), Statistical Process Control

PROJECTS

Software Safety Analysis for Medical Device Application

- Conducted comprehensive hazard analysis and risk assessments for a simulated medical device software application, identifying potential safety hazards and contributing to risk mitigation strategies.
- Mapped design verification and validation activities to risk management documentation, ensuring traceability and compliance with **IEC 62304** standards for software lifecycle processes.
- Documented findings and recommendations in a structured risk management file, incorporating severity and occurrence assessments for potential failure modes.

Reliability Study Design for Enterprise System Component

- Designed experiments (DOE) for a critical enterprise system component to evaluate the impact of varying environmental factors on performance and reliability.
- Analyzed experimental data using **Python (pandas, NumPy)** and statistical methods to identify key drivers of system degradation and propose improvements.
- Developed an initial draft of a reliability program plan, outlining testing methodologies and data collection requirements for future product iterations.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)